Sara Busatto, Ph.D.

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WORK EXPERIENCE

11/2024 – Present **Tenure Track Assistant Professor**
Biological Nanoparticles Laboratory
Nanomedicine and Drug Delivery Department
Groningen Research Institute of Pharmacy
Faculty of Science and Engineering
University of Groningen
Groningen, The Netherlands

POSTDOCTORAL TRAINING, EDUCATION AND POSITIONS

06/2020 – 10/2024 **Postdoctoral Fellow**
Moses Laboratory
Vascular Biology Program
Boston Children's Hospital, Harvard Medical School
Boston, Massachusetts, United States
Advised by Prof. Marsha Moses

06/2019 – 05/2020 **Postdoctoral Fellow**
Nanomedicine and Extracellular Vesicles Laboratory
Mayo Clinic, Jacksonville; Florida, United States
Advised by Prof. Joy Wolfram

11/2015 – 03/2019 **Ph.D. in Health Technology**
Colloidal Clinical Chemistry Laboratory
University of Brescia, Brescia, Italy
Advised by Prof. Paolo Bergese and Prof. Marco Rusnati
Visiting Ph.D. Student (2018) (http://www.mayo.edu/research/labs/nanomedicine-extracellular-vesicles/about/research-team)
Nanomedicine and Extracellular Vesicles Laboratory
Mayo Clinic, Jacksonville; Florida, United States
Advised by Prof. Joy Wolfram

10/2013 – 10/2015 **M.S. in Medical Biotechnology**
Department of Molecular and Translational Medicine
University of Brescia, Brescia, Italy
Advised by Prof. Paolo Bergese

08/2012 – 09/2013 **Laboratory Technician**
Oxy-Gen Laboratory, Brescia, Italy

10/2008 – 11/2011 **B.S. in Biomedicine and Clinical Science**
Faculty of Medicine
University of Trieste, Trieste, Italy
Advised by Prof. Rossana Bussani

AWARDS

250 USD	Best Poster Award Brain-seeking extracellular vesicles derived from metastatic breast cancer cells modulate the metabolism of the endothelial cells of the blood-brain barrier Dana Farber Breast & Gynecologic Cancer Symposium (4/2023)
Travel Expenses	Invited Lecturer at XXVII School of Pure and Applied Biophysics Conference Expenses and Accommodation. Lipoproteins as key components of the Tumor-derived Extracellular vesicles interactome XXVII School of Pure and Applied Biophysics (2/2023)
1,200 USD	AACR-Breast Cancer Research Foundation Scholar-in-Training Award Conference Expenses. Brain-seeking extracellular vesicles derived from metastatic breast cancer cells modulate brain endothelial cell metabolism. AACR Annual Meeting, New Orleans (4/2022).
700 USD	Conference Expenses (airfare, and accommodation). Brain-Tropic Extracellular Vesicles for Treatment of Brain Tumors. New York Academy of Sciences STEM Exchange: Research and Career Symposium, New York, United States (8/2018)
Travel expenses	Conference Expenses (registration fee, airfare, and accommodation). Tangential flow filtration-based isolation of extracellular vesicles for biomedical applications. Repligen Inspiring Advances in Bioprocessing. What's new in single-use, continuous and flexible bioprocessing, industry, case studies. Durham, North Carolina, United States (6/2018).
Travel expenses	Conference Expenses (registration fee, airfare, and accommodation) Synthetic and biological nanotherapeutics. Mayo Clinic, Arizona, Science of Medicine Seminar Series. Mayo Clinic, Phoenix, Arizona, United States (6/2018)
Travel expenses	Conference Fellowship (registration fees, airfare, and accommodation) 11th European and Global Summit for Clinical Nanomedicine, Targeted Delivery and Precision Medicine, CLINAM, Basel, Switzerland (9/2018)
Travel expenses	Conference Fellowship (registration fees, airfare, and accommodation) GEIVEX Group, Barcelona, Spain (3/2017)
Travel expenses	Conference fellowship (registration fees, airfare and accommodation) National Interuniversity Consortium of Materials Science and Technology (INSTM). Firenze, Italy (7/2016)
1,000 €	Graduation Award Friulovest Banca, Spilimbergo, Italy (5/2016)

FUNDING

168,000 USD	**Trainee (Degree Level: Post-doc)** T32 Institutional Training Program (5T32HL007917-22) Prof. Robert C. Flaumenhaft. Beth Israel Deaconess Medical Center, Harvard Medical School, United States (8/2021 – Present). Advised by Prof. Marsha Moses.
4,320 USD	**Visiting Ph.D. Student Fellowship** Nanomedicine and Extracellular Vesicles Laboratory, Mayo Clinic, Jacksonville, Florida, United States (1-9/2018)
5,000 €	**Visiting Ph.D. Student Fellowship** Italian Ministry of Education and University of Brescia, Brescia, Italy (1-9/2018)
40,000 €	**Ph.D. Fellowship** Italian Ministry of Education and University of Brescia, Brescia, Italy (11/2015-11/2018)
1,500 €	**Thesis Abroad Project**

Italian Ministry of Education and University of Brescia, Brescia, Italy (3-6/2015)

LEADERSHIP SKILLS

2025-Present — Leading member of the Local Organization Committee of the Annua Meeting of the Netherlands Society of Extracellular Vesicles sponsored by the Netherlands Society of Extracellular Vesicles to be hosted in Groningen, The Netherlands – **Organizer.**

2024-2025 — Served as a member of the International Organization Committee of the Inaugural Workshop on the extracellular vesicles biomolecular corona sponsored by the International Society of Extracellular Vesicles held in Brescia, Italy– **Organizer.**

2023-2024 — Co-organizer of the inaugural Boston Children's Hospital (BCH) Postdoc Symposium in collaboration with the Senior Vice President of Research Administration and the Equality Diversity and Inclusion committee. The event will engage with scientists working in the Boston Longwood Medical Area and focus on the mentoring of postdocs, with a particular emphasis on those coming from underrepresented communities. Boston, United States – **Organizer.**

2021-2024 — Organizer of the Annual Young Minds Science Review Event in collaboration with the journal Frontiers for Young Minds. Distinguished scientists working at Boston Children's Hospital were invited to write about their cutting-edge discoveries in a language accessible for young readers. Kids themselves (8-15 years old) – with the help of a science mentor (postdoc working at Boston Children's Hospital) read the paper and, during a virtual event, provided feedback and explained to the authors how to best improve the articles before publication. Boston, United States – **Organizer.**

2021-2022 — Co-Organizer of the annual All-Star Mentoring Event in collaboration with Boston Children's Hospital Postdoc Association. Postdocs working at Boston Children's Hospital or other hospitals in the Longwood area get the chance to meet and talk with outstanding scientists and mentors working in Biotech companies, regulatory agencies, or academia. Boston, United States – **Organizer.**

2020 - present — Boston Children's Hospital Postdoc Association Mentoring Committee. Boston, United States – **Co-chair.**

2020 - present — Boston Children's Hospital Green Labs Recycling Committee – **Co-chair.**

07/2019 — Leader Launch Program Zoom Cohort. Program designed to create higher levels of confidence and effectiveness in leadership organized by Mayo Clinic Office of Entrepreneurship, Office of Research Diversity and Inclusion and Center for Biomedical Discovery Partnering with Capita 3. Mayo Clinic, Jacksonville, Florida, United States – **Attendee.**

12/2018 — Colorimetric Nanoplasmonic Assay Workshop, Utrecht, Netherlands – **Workshop coordinator.**

TEACHING

1. University of Groningen (11/2024 – Present) – **Lecturer.**

2. XXVII School of Pure and Applied Biophysics. Venice Institute for Arts and Sciences, Venice, Italy (2/2023) – **Invited Lecturer.**

3. Basic Research Internship in Neuroscience and Cancer (BRINC). Mayo Clinic, Jacksonville, Florida, United States and University of North Florida (8/2019 – 5/2020) – **Collaborator, teacher, and student monthly meeting organizer.**

4. Mayo Clinic, Bolles Science and Medicine Camp. Mayo Clinic, Jacksonville, Florida, United States (7/2019) – **Event Organizer.**

5. Mayo Clinic Girl Scouts of Gateway Council Science Boot Camp, Mayo Clinic, Jacksonville, Florida, United States (3/2018) – **Event Organizer.**

6. High School Science Visit, Mandarin High School, Jacksonville, Florida, United State (4/2018) – **Event Organizer.**

7. Chemistry and biology, Brescia, Italy (2015 - present) – **Tutor.**

MENTORING

Mentees:

Sjoerd Idzerda	Ph.D. Student, Biological Nanoparticles Laboratory, Department of Nanomedicine and Drug Targeting, Groningen Research Institute of Pharmacy, University of Groningen, The Netherlands (01/2025 – Present).
Marloes Kruk	Research Assistant, Biological Nanoparticles Laboratory, Department of Nanomedicine and Drug Targeting, Groningen Research Institute of Pharmacy, University of Groningen, The Netherlands (11/2024 – Present).
Laura Sikkens	MSc. Student, Biological Nanoparticles Laboratory, Department of Nanomedicine and Drug Targeting, Groningen Research Institute of Pharmacy, University of Groningen, The Netherlands (01/2026 – Present).
Sterre Metz	MSc. Student, Biological Nanoparticles Laboratory, Department of Nanomedicine and Drug Targeting, Groningen Research Institute of Pharmacy, University of Groningen, The Netherlands (01/2026 – Present).
Kaj Mulder	MSc. Student, Biological Nanoparticles Laboratory, Department of Nanomedicine and Drug Targeting, Groningen Research Institute of Pharmacy, University of Groningen, The Netherlands (01/2026 – Present).
Stephen Cobbs	Research Assistant, Moses Laboratory, Vascular Biology Program. Boston Children's Hospital (06/2023 – present), Boston, Massachusetts, United States. Current position: Research Assistant, Boston Children's Hospital, Boston, MA, USA.
Rama Aldakhlallah	Research Assistant, Moses Laboratory, Vascular Biology Program. Boston Children's Hospital (06/2020 – 07/2023), Boston, Massachusetts, United States. Current position: Ph.D. student at Scripps Health Institute, San Diego, CA, USA. **Outcome:** two co-author publications.
Brandon Griggs	Student at Robert E. Lee High School Jacksonville, Florida, United States. Spark Research Mentorship Program, student intern (06/2019 – 01/2020), Mayo Clinic, Jacksonville, Florida, United States. Current position: Undergraduate Student (Yale University, Connecticut, United States).
Sierra Walker	Ph.D. student in Biochemistry and Molecular Biology Mayo Clinic, Jacksonville, Florida, United States. Current position: Ph.D. student (01/2019 – present) Mayo Clinic, Jacksonville, Florida, United States. **Outcomes:** three first-author and several co-author publications.
Miriam Romano	Ph.D. student in Precision Medicine, University of Brescia, Brescia, Italy. Current position: Ph.D. student (11/2018 – present), University of Brescia, Brescia, Italy. **Outcome:** one first-author publication.
Taylor Ticer	Undergraduate Student in Biology, University of North Florida at Jacksonville, Florida, United States. Student intern (10/2017 – 08/2018), Mayo Clinic, Jacksonville, Florida, United States. Current position: Ph.D. student at Medical University of South Carolina, Charleston, South Carolina, United States. **Outcomes:** two co-author publications.
Simone Zaghen	Undergraduate Student in Biotechnology, University of Brescia, Brescia, Italy. Student intern (09/2017 – 07/2018), University of Brescia, Brescia, Italy. Current position: Ph.D. Student at Chalmers Tekniska Hogskola, Sweden.

PUBLICATIONS (https://www.scopus.com/authid/detail.uri?authorId=57188690042)

1. J. Sesen#, T. Martinez#, **S. Busatto#**, L. Poluben, H. Nassour, C. Stone, K. Ashok, M. A. Moses, E. R. Smith and A. Ghalali. AZIN1 level is increased in medulloblastoma and correlates with c-Myc activity and tumor phenotype. Journal of *Experimental and Clinical Cancer Research* 44, 56, 2025. #co-first authors. https://doi.org/10.1186/s13046-025-03274-1

2. **S. Busatto**, T. H. Song, H. J. Kim, C. Hallinan, M. N. Lombardo, A. O. Stemmer-Rachamimov, K. Lee, M. A. Moses. Breast Cancer-Derived Extracellular Vesicles Modulate the Cytoplasmic and Cytoskeletal Dynamics of Blood-Brain Barrier Endothelial Cells. *Journal of Extracellular Vesicles* 14(1), 2025. https://doi.org/10.1002/jev2.70038

3. J.A. Welsh… **S. Busatto**… H. Zubair. Minimal information for studies of extracellular vesicles (MISEV2023): From basic to advanced approaches. Journal of Extracellular Vesicles, 2024. https://doi.org/10.1002/jev2.12404

4. J. Pendiuk Goncalves, S. A. Walker, J. S. Aguilar Díaz de León, Y. Yang, I. Davidovich, **S. Busatto**, J. Sarkaria, Y. Talmon, C. R. Borges, J. Wolfram. Glycan Node Analysis Detects Varying Glycosaminoglycan Levels in Melanoma-Derived Extracellular Vesicles. *International Journal of Molecular Sciences*, 2023. https://doi.org/10.3390/ijms24108506.

5. S. A. Walker**,** I. Davidovich, Y. Yang, A. Lai, J. Pendiuk Goncalves, V. Deliwala, **S. Busatto**, S. Shapiro, N. Koifman, C. Salomon, Y. Talmon, J. Wolfram. Sucrose-based cryoprotective storage of extracellular vesicles. *Extracellular Vesicle*, 2022. https://doi.org/10.1016/j.vesic.2022.100016.

6. **S. Busatto***, Y. Yang, D. Iannotta, I. Davidovich, Y. Talmon, J. Wolfram*. Considerations for extracellular vesicle and lipoprotein interactions in cell culture assays. *Journal of Extracellular Vesicles*, 2022; https://doi.org/10.1002/jev2.12202. *Corresponding authors

7. **S. Busatto**, G. Morad, P. Guo, M. Moses. The role of extracellular vesicles in the physiological and pathological regulation of the blood-brain barrier. *FASEB BioAdvances*, 2021; 3(9):665-675. doi: 10.1096/(ISSN)2573-9832. **Cover of the Volume 3, Issue 9, September 2021 https://doi.org/10.1096/fba.1139.**

8. **S. Busatto***, D. Iannotta, L. Di Marzio, J. Wolfram*. A simple and quick method for loading proteins in extracellular vesicles. *Pharmaceuticals*, 2021; 14(4):356. doi: 10.3390/ph14040356. *Corresponding authors

9. J. L. Ruiz, J. D. Hutcheson, L. Cardoso, T. Pham, **S. Busatto**, S. Federici, A. Ridolfi, P. Bergese, S. Weinbaum, E. Aikawa. Nanoanalytical Analysis of Bisphosphonate-Induced Alterations of Microcalcifications using a 3D Hydrogel Platform. *Proceedings of the National Academy of Sciences of the United States of America,* 2020; 118(14):e1811725118. doi: 10.1073/pnas.1811725118.

10. P. Guo, **S. Busatto**, J. Huang, G. Morad, M. A. Moses. A facile magnetic extrusion method for preparing endosome-derived vesicles for cancer drug delivery. *Advanced Functional Materials*, 2020; PMC Journal – In Press. doi: 10.1002/adfm.202008326. **Cover of the Volume 31, Number 44, October 2021 https://doi.org/10.1002/adfm.202170328.**

11. **S. Busatto***, Y. Yang, S. A. Walker, I. Davidovich, W. H. Lin, L. Lewis-Tuffin, P. Z. Anastasiadis, J. Sarkaria, Y. Talmon, G. Wurtz, J. Wolfram*. Brain metastases-derived extracellular vesicles induce binding and aggregation of low-density lipoprotein. *Journal of Nanobiotechnology*, 2020; 18(1):162. doi: 10.1186/s12951-020-00722-2. *Corresponding authors

12. S. A. Walker, J. S. Aguilar Díaz De león, **S. Busatto**, G. A. Wurtz, A. C. Zubair, C. R. Borges, J. Wolfram. Glycan node analysis of plasma-derived extracellular vesicles. *Cells*, 2020; 9(9):1946. doi: 10.3390/cells9091946.

13. **S. Busatto***, S. Walker, W. Grayson, A. Pham, M. Tian, N. Nesto, J. Barklund, J. Wolfram*. Lipoprotein-based drug delivery. *Advanced Drug Delivery Reviews,* 2020; 159:377-390. doi: 10.1016/j.addr.2020.08.003. *Corresponding authors

14. M. Romano, A. Zendrini, L. Paolini, **S. Busatto**, A. C. Berardi, P. Bergese and A. Radeghieri. Extracellular vesicles in regenerative medicine. Book Chapter. Nanomaterials for theranostics and tissue engineering. *Elsevier Micro & Nano Technologies*, 2020; 29-58. doi: 10.1016/B978-0-12-817838-6.00002-4.

15. A. Zendrini, L. Paolini, **S. Busatto**, A. Radeghieri, M.H. Wauben, M. van Herwijnen, P. Nejsum, A. Borup, A. Ridolfi, C. Montis, P. Bergese. Purity and concentration of microliter formulations of EVs by an augmented COlloidal NANoplasmonic (CONAN) method. *Frontiers in Bioengineering and Biotechnology*, 2019; 7:452. doi: 10.3389/fbioe.2019.00452.

16. **S. Busatto,** A.Zendrini, M. Romano, L. Paolini, A. Radeghieri, M. Presta, P. Bergese. The nanostructured secretome. *Biomaterials Science*, 2019; 8(1):39-63. doi: 10.1039/C9BM01007F**.**

17. M. Tian, T. Ticer, Q. Wang, S. Walker, A. Pham, A. Suh, **S. Busatto**, I. Davidovich, R. Alkharboosh, A. Quinones-Hinojosa, Y.Talmon, S. Shapiro, F. Rückert, J. Wolfram. Anti-inflammatory biogenic adipose nanoparticles for combination therapy. *Small*, 2019; 16(10):e1904064. doi: 10.1002/smll.201904064.

18. **S. Walker**[#]**, S. Busatto**[#], A. Pham, M. Tian, A. Suh, K. Carson, A. Quintero, M. Lafrence, H. Malik, M. X. Santana, J. Wolfram. Extracellular vesicle-based drug delivery systems for cancer treatment. *Theranostics*, 2019; 9(26):8001-8017. doi: 10.7150/thno.37097. #Equal contribution

19. H. S. Leong, K. S. Butler, C J. Brinker, M. Azzawi, S. Conlan, C. Dufès, A. Owen, S. Rannard, C. Scott, C. Chen, M. A. Dobrovolskaia, S. V. Kozlov, A. Prina-Mello, R. Schmid, P. Wick, F. Caputo, P. Boisseau, R. M. Crist, S. E. McNeil, B. Fadeel, L. Tran, S. Foss Hansen, N. B. Hartmann, L. P. W. Clausen, L. M. Skjolding, A. Baun, M. Ågerstrand, Z. Gu, D. A. Lamprou, C. Hoskins, L. Huang, W. Song, H. Cao, X. Liu, K. D. Jandt, W. Jiang, B. Y. S. Kim, K. E. Wheeler, A. J. Chetwynd, I. Lynch, S. Moein Moghimi, A. Nel, T. Xia, P. S. Weiss, B. Sarmento, J. das Neves, H. A Santos, L. Santos, S. Mitragotri, S. Little, D. Peer, M.M. Amiji, M. José Alonso, A. Petri-Fink, S. Balog, A. Lee, B. Drasler, B. Rothen-Rutishauser, S. Wilhelm, H. Acar, R. G. Harrison, C. Mao, P. Mukherjee, R. Ramesh, L. R. McNally, **S. Busatto**, J. Wolfram, P. Bergese, M. Ferrari, R. H Fang, L. Zhang, J. Zheng, C. Peng, B. Du, M. Yu, D. M. Charron, G. Zheng, C. Pastore. On the issue of transparency and reproducibility in nanomedicine. *Nature Nanotechnology*, 2019; 14(7):629-635. doi: 10.1038/s41565-019-0496-9.

20. **S. Busatto***, A. Pham, A. Suh and J. Wolfram*. Organotropic drug delivery: synthetic nanoparticles and extracellular vesicles. *Biomedical Microdevices*, 2019; 21(2):46. doi: 10.1007/s10544-019-0396-7. *Corresponding authors

21. A. Salvi, M. Vezzoli, **S. Busatto**, L. Paolini, T. Faranda, E. Abeni, M. Caracausi, F. Antonaros, A. Piovesan, C. Locatelli, G. Cocchi, G. Alvisi, G. De Petro, D. Ricotta, P. Bergese and A. Radeghieri. Analysis of a Nanoparticle-enriched fraction of plasma reveals miRNA candidates for DS pathogenesis. *International Journal of Molecular Medicine*, 2019; 43(6):2303-2318. doi: 10.3892/ijmm.2019.4158.

22. **S. Busatto***, G. Villanilam, T. Ticer, W. L. Lin, D. W. Dickson, A. E. Thompson, S. Shapiro, P. Bergese* and J. Wolfram*. Tangential Flow Filtration for Highly Efficient Concentration of Extracellular Vesicles from Large Volumes of Fluid. *Cells*, 2018; 7(12):273. doi:10.3390/cells7120273. *Corresponding authors

23. C. Théry, K.W. Witwer, E. Aikawa, M.J. Alcaraz, J.D. Anderson, R. Andriantsitohaina, A. Antoniou, T. Arab, F. Archer, G.K. Atkin-Smith, D.C. Ayre, J.M. Bach, D. Bachurski, H. Baharvand, L. Balaj, S. Baldacchino, N.N. Bauer, A.A. Baxter, M. Bebawy, C. Beckham, A. Bedina Zavec, A. Benmoussa, A.C. Berardi, P. Bergese, E. Bielska, C. Blenkiron, S. Bobis-Wozowicz, E. Boilard, W. Boireau, A. Bongiovanni, F.E. Borràs, S. Bosch, C.M. Boulanger, X. Breakefield, A.M. Breglio, M.Á. Brennan, D.R. Brigstock, A. Brisson, M.L. Broekman, J.F. Bromberg, P. Bryl-Górecka, S. Buch, A.H. Buck, D. Burger, **S. Busatto**, D. Buschmann, B. Bussolati, E.I. Buzás, J.B. Byrd, G. Camussi, D.R. Carter, S. Caruso, Chamley, Chang YT, Chen C, Chen S, Cheng L, Chin AR, Clayton A, Clerici SP, Cocks A, Cocucci E, Coffey RJ, L.W. Cordeiro-da-A. Silva, Y. Couch, F.A. Coumans, B. Coyle, R. Crescitelli, M.F. Criado, C. D'Souza-Schorey, S. Das, Datta A. Chaudhuri, P. de Candia, E.F. De Santana, O. De Wever, H.A. Del Portillo, T. Demaret, S. Deville, A. Devitt, B. Dhondt, D. Di Vizio, L.C. Dieterich, V. Dolo, A.P. Dominguez Rubio, M. Dominici, M.R. Dourado, T.A. Driedonks, F.V. Duarte, H.M. Duncan, R.M. Eichenberger, K. Ekström, S. El Andaloussi, C. Elie-Caille, U. Erdbrügger, J.M. Falcón-Pérez, F. Fatima, J.E. Fish, M. Flores-Bellver, A. Försönits, A. Frelet-Barrand, F. Fricke, G. Fuhrmann, S. Gabrielsson, A. Gámez-Valero, C. Gardiner, K. Gärtner, R. Gaudin, Y.S. Gho, B. Giebel, C. Gilbert, M. Gimona, I. Giusti, D.C. Goberdhan, A. Görgens, S.M. Gorski, D.W. Greening, J.C. Gross, A. Gualerzi, G.N. Gupta, D. Gustafson, A. Handberg, R.A. Haraszti, P. Harrison, H. Hegyesi, A. Hendrix, A.F. Hill, F.H. Hochberg, K.F. Hoffmann, B. Holder, H. Holthofer, B. Hosseinkhani, G. Hu, Y. Huang, V. Huber, S. Hunt, A.G. Ibrahim, T. Ikezu, J.M. Inal, M. Isin, A. Ivanova, H.K. Jackson, S. Jacobsen, S.M. Jay, M. Jayachandran, G. Jenster, L. Jiang, S.M. Johnson, J.C. Jones, A. Jong, T. Jovanovic-Talisman, S. Jung, R.

Kalluri, S.I. Kano, S. Kaur, Y. Kawamura, E.T. Keller, D. Khamari, E. Khomyakova, A. Khvorova, P. Kierulf, K.P. Kim, T. Kislinger, M. Klingeborn, D.J. 2nd Klinke, M. Kornek, M.M. Kosanović, Á.F. Kovács, E.M. Krämer-Albers, S. Krasemann, M. Krause, I.V. Kurochkin, G.D. Kusuma, S. Kuypers, S. Laitinen, S.M. Langevin, L.R. Languino, J. Lannigan, C. Lässer, L.C. Laurent, G. Lavieu, E. Lázaro-Ibáñez, S. Le Lay, M.S. Lee, Y.X.F. Lee, D.S. Lemos, M. Lenassi, A. Leszczynska, I.T. Li, K. Liao, S.F. Libregts, E. Ligeti, R. Lim, S.K. Lim, A. Linē, K. Linnemannstöns, A. Llorente, C.A. Lombard, M.J. Lorenowicz, Á.M. Lörincz, J. Lötvall, J. Lovett, M.C. Lowry, X. Loyer, Q. Lu, B. Lukomska, T.R. Lunavat, S.L. Maas, H. Malhi, A. Marcilla, J. Mariani, J. Mariscal, E.S. Martens-Uzunova, L. Martin-Jaular, M.C. Martinez, V.R. Martins, M. Mathieu, S. Mathivanan, M. Maugeri, L.K. McGinnis, M.J. McVey, D.G. Jr Meckes, K.L. Meehan, I. Mertens, V.R. Minciacchi, A. Möller, M. Møller Jørgensen, A. Morales-Kastresana, J. Morhayim, F. Mullier, M. Muraca, L. Musante, V. Mussack, D.C. Muth, K.H. Myburgh, T. Najrana, M. Nawaz, I. Nazarenko, P. Nejsum, C. Neri, T. Neri, R. Nieuwland, L. Nimrichter, J.P. Nolan, E.N. Nolte-'t Hoen, N. Noren Hooten, L. O'Driscoll, T. O'Grady, A. O'Loghlen, T. Ochiya, M. Olivier, A. Ortiz, L.A. Ortiz, X. Osteikoetxea, O. Østergaard, M. Ostrowski, J. Park, D.M. Pegtel, H. Peinado, F. Perut, M.W. Pfaffl, D.G. Phinney, B.C. Pieters, R.C. Pink, D.S. Pisetsky, E. Pogge von Strandmann, I. Polakovicova, I.K. Poon, B.H. Powell, I. Prada, L. Pulliam, P. Quesenberry, A. Radeghieri, R.L. Raffai, S. Raimondo, J. Rak, M.I. Ramirez, K.M. G. Raposo, M.S. Rayyan, N. Regev-Rudzki, F.L. Ricklefs, P.D. Robbins, D.D. Roberts, S.C. Rodrigues, E. Rohde, S. Rome, Rouschop, A. Rughetti, A.E. Russell, P. Saá, S. Sahoo, E. Salas-Huenuleo, C. Sánchez, J.A. Saugstad, M.J. Saul, R.M. Schiffelers, R. Schneider, T.H. Schøyen, A. Scott, E. Shahaj, S. Sharma, O. Shatnyeva, F. Shekari, G.V. Shelke, A.K. Shetty, K. Shiba, P.R. Siljander, A.M. Silva, A. Skowronek, O.L. 2nd Snyder, R.P. Soares, B.W. Sódar, C. Soekmadji, J. Sotillo, P.D. Stahl, W. Stoorvogel, S.L. Stott, E.F. Strasser, S. Swift, H. Tahara, M. Tewari, K. Timms, S. Tiwari, R. Tixeira, M. Tkach, W.S. Toh, R. Tomasini, A.C. Torrecilhas, J.P. Tosar, V. Toxavidis, L. Urbanelli, P. Vader, B.W. van Balkom, S.G. van der Grein, J. Van Deun, M.J. van Herwijnen, K. Van Keuren-Jensen, G. van Niel, M.E. van Royen, A.J. van Wijnen, M.H. Vasconcelos, I.J. Jr. Vechetti, T.D. Veit, L.J. Vella, É. Velot, F.J. Verweij, B. Vestad, J.L. Viñas, T. Visnovitz, K.V. Vukman, J. Wahlgren, D.C. Watson, M.H. Wauben, A. Weaver, J.P. Webber, V. Weber, A.M. Wehman, D.J. Weiss, J.A. Welsh, S. Wendt, A.M. Wheelock, Z. Wiener, L. Witte, J. Wolfram, A. Xagorari, P. Xander, J. Xu, X. Yan, M. Yáñez-Mó, H. Yin, Y. Yuana, V. Zappulli, J. Zarubova, V. Žėkas, J.Y. Zhang, Z. Zhao, L. Zheng, A.R. Zheutlin, A.M. Zickler, P. Zimmermann, A.M. Zivkovic, D. Zocco, E.K. Zuba-Surma. Minimal Information for Studies of Extracellular Vesicles 2018 (MISEV2018): a position statement of the International Society for Extracellular Vesicles and update of the MISEV2014 guidelines. _Journal of Extracellular Vesicles_, 2018; 7(1):1535750. doi: 10.1080/20013078.2018.1535750.

24. J. Pelt[#], **S. Busatto*[#]**, M. Ferrari, E. A. Thompson, K Mody and J. Wolfram. Chloroquine and nanoparticle drug delivery: A promising combination. _Pharmacology and Therapeutics_, 2018; 191:43-49. doi: 10.1016/j.pharmthera.2018.06.007. [#]Equal contribution, *Corresponding authors

25. **S. Busatto**, A. Giacomini, C. Montis, R. Ronca and P. Bergese. Uptake profiles of human serum exosomes by murine and human tumor cells through combined use of colloidal nanoplasmonics and flow cytofluorimetric analysis. _Analytical Chemistry_, 2018; 90(13):7855-7861. doi: 10.1021/acs.analchem.7b04374.

26. V. Salvi, V. Gianello, **S. Busatto**, P. Bergese, L. Andreoli, U. D'Oro, A. Zingoni, A. Tincani, D. Bosisio and S. Sozzani. Exosome-delivered microRNAs promote IFN-α secretion by plasmacytoid DCs via TLR7. _Journal of Clinical Investigation Insight_, 2018; 3(10):e98204. doi: 10.1172/jci.insight.98204.

27. C. Montis[#], **S. Busatto**[#], F. Valle, A. Zendrini, A. Salvatore, Y. Gerelli, D. Berti, P. Bergese. Biogenic supported lipid bilayers from nanosized extracellular vesicles. _Advanced Biosystems_, 2018; 2, 1700200. doi: 10.1002/adbi.201700200. [#]Equal contribution

28. L. Paolini, F. Orizio, **S. Busatto**, A. Radeghieri, R. Bresciani, P. Bergese, and E. Monti. Exosomes Secreted by HeLa Cells Shuttle on Their Surface the Plasma Membrane-Associated Sialidase NEU3. _Biochemistry_ 2017; 56(48):L6401-6408. doi: 10.1021/acs.biochem.7b00665.

29. M. Berardocco, A. Radeghieri, **S. Busatto**, M. Gallorini, C. Raggi, C. Gissi, I. D'Agnano, P. Bergese, A. Felsani, A. C. Berardi. RNA-seq reveals distinctive RNA profiles of extracellular vesicles from different human liver cancer cell lines. _Oncotarget_, 2017; 8(47):82920-82939. doi: 10.18632/oncotarget.20503.

30. C. Montis, A. Zendrini, F. Valle, **S. Busatto**, L. Paolini, A. Radeghieri, A. Salvatore, D. Berti and P. Bergese. Size Distribution of Extracellular Vesicles by Optical Correlation Techniques. _Colloids and Surfaces B: Biointerfaces_, 2017; 158:331–338. doi: 10.1016/j.colsurfb.2017.06.047.

31. L. Paolini, A. Zendrini, G. Di Noto, **S. Busatto**, E. Lottini, A. Dossi, A. Caneschi, D. Ricotta and P. Bergese. Residual matrix from different separation techniques impacts exosomes biological activity. *Scientific Reports*, 2016; 6:23550. doi: 10.1038/srep23550.

PATENTS

S. Busatto, M.A. Moses, et al. Machine learning methods for determining cell morphology and motility. Provisional Application Filing C1233.70300US00, United States.

CONFERENCES

Invited speaker

1. **S. Busatto**. Extracellular Vesicles and Lipoproteins: interacting nanoparticles with a potential role in human pathogenesis. 1st ISEV Workshop on Extracellular Vesicle Biomolecular Corona. Brescia, Italy (02/2025) – **Invited Lecturer.**

2. **S. Busatto**. Lipoproteins as key components of the Tumor-derived Extracellular vesicles interactome. XXVII School of Pure and Applied Biophysics 2023. Venice, Italy (02/2023) – **Invited Lecturer.**

3. **S. Busatto**, G. Morad, M.A. Moses. Breast metastasis-derived extracellular vesicles modulate the metabolism of multiple blood-brain barrier components. International Caparica Conference on Analytical Proteomics 2022. Caparica, Portugal (07/2022) – **Invited Speaker.**

4. **S. Busatto**. 2nd EV Journal Club. Consideration for extracellular vesicle and lipoprotein interactions in cell culture assays. EV Journal club, International Society of Extracellular Vesicles (05/2022) – **Invited webinar Speaker** (https://youtu.be/gUgL-omLBEI).

5. **S. Busatto.** Brain metastases-derived extracellular vesicles induce binding and aggregation of low-density lipoproteins. EV Journal club, Witwer Laboratory at Johns Hopkins School of Medicine, Baltimore, United States (03/2021) – **Invited webinar Speaker** (https://youtu.be/kM7J1PofqB8)**.**

6. **S. Busatto.** Extracellular vesicles and the nanostructured secretome. Webinar in Drug Research, University of Helsinki, Finland (11/2020) – **Invited Speaker.**

7. **S. Busatto**. Extracellular vesicles as new tools in nanomedicine. EPFL Group Seminar. École Polytechnique Fèdèrale de Lausanne, Lausanne, Switzerland (02/2019) – **Invited Speaker.**

8. **S. Busatto**. Cancer cell-derived extracellular vesicles as diagnostic and therapeutic tools. Gibco Cell Culture Hero Webinar, (Global 11/2018) – **Invited webinar Speaker** (https://www.labroots.com/ms/webinar/extracellular-vesicles-diagnostic-therapeutic-tools-cancer)**.**

9. **S. Busatto**. Tangential Flow Filtration-based Isolation of Extracellular Vesicles for Biomedical Applications. Repligen Inspiring Advances in Bioprocessing (What's new in single-use, continuous and flexible bioprocessing, industry case studies), Durham, North Carolina, United States (6/2018) – **Invited speaker.**

10. **S. Busatto**, Paolo Bergese. Isolation and characterization of extracellular vesicles obtained from different biological fluids. GEIVEX Workshop, Barcelona, Spain (3/2017) – **Invited Speaker.**

Speaker

1. **S. Busatto**, G. Morad, M.A. Moses. Breast cancer-derived extracellular vesicles modulate the endocytic metabolism and morphodynamics of the endothelial cells of the blood-brain barrier. Dana Farber and Harvard Cancer Center Symposium on Breast and Gynecological Cancers. Boston, United States (03/2024) – **Speaker.**

2. **S. Busatto**, G. Morad, M.A. Moses. Brain-seeking extracellular vesicles derived from metastatic breast cancer cells modulate brain endothelial cell metabolism. American Association for Cancer Research Annual Meeting 2022. New Orleans, United States (04/2022) – **Speaker.**

3. **S. Busatto.** The double-edged sword of extracellular vesicles: key players in cancer metastasis and promising drug-delivery nanovehicles for the treatment of cancer and its metastasis. Work in Progress Seminar. Vascular Biology Program, Boston Children's Hospital, Boston United, States. (12/2021) **- Speaker**.
4. **S. Busatto.** Exploring the interface between cancer cell-derived extracellular vesicles and the endothelium. Work in progress seminars. Mayo Clinic, Jacksonville, Florida, United States (07/2019) – **Speaker.**

5. **S. Busatto,** J. Wolfram. Synthetic and biological nanoparticles for therapeutic applications. Mayo Clinic Arizona Science of Medicine Seminar Series. Mayo Clinic, Phoenix, Arizona, United States (6/2018) – **Speaker.**

6. **S. Busatto**. Medical applications of extracellular vesicles. Mayo Clinic Extracellular Vesicles Interest Meeting. Mayo Clinic, Rochester/ Jacksonville, Florida, United States (3/2018) – **Speaker.**

7. **S. Busatto**, C. Montis, A. Zendrini, D. Berti, A. Salvatore, F. Valle, Y. Gerelli, D. Berti, P. Bergese. The first extracellular vesicle-made supported lipid bilayer biosensor. 11th European and Global Summit for Clinical Nanomedicine, Targeted Delivery and Precision Medicine, Basel, Switzerland (9/2018) – **Speaker.**

8. **S. Busatto,** T. Ticer, N. Shukla, J. Wolfram. Nanomedicine. Jacksonville University Science and Engineering Lecture Series, Jacksonville, Florida, United States (3/2018) – **Speaker.**

9. **S. Busatto.** Medical applications of extracellular vesicles. Work in Progress Seminar series, Mayo Clinic. Jacksonville, Florida, United States (3/2018) – **Speaker.**

10. **S. Busatto**. Extracellular vesicles, a new perspective for nanomedicine. International School of Nanomedicine, Erice, Italy (4/2017) – **Speaker.**

11. **S. Busatto**. Nanoscale characterization of extracellular vesicle is mandatory to assess their biological activity and biotechnological exploitation. INSTM National Young Researcher Forum on Material Science and Technology, Ischia Terme, Italy (7/ 2016) – **Speaker.**

Poster presentations

1. **S. Busatto**, G. Morad, M.A. Moses. Breast cancer-derived extracellular vesicles modulate the endocytic metabolism and morphodynamics of the endothelial cells of the blood-brain barrier. American Association for Cancer Research Annual Meeting. San Diego, United States (04/2024) – **Poster.**

2. **S. Busatto**, G. Morad, M. A. Moses. Brain-seeking extracellular vesicles derived from metastatic breast cancer cells modulate the metabolism of the endothelial cells of the blood-brain barrier. Judah Folkman Research Day 2023. Boston Children's Hospital, Boston, United States (05/2023) – **Poster.**

3. **S. Busatto**, G. Morad, M. A. Moses. Brain-seeking extracellular vesicles derived from metastatic breast cancer cells modulate the metabolism of the endothelial cells of the blood-brain barrier. Dana Farber Breast & Gynecologic Cancer Symposium. Harvard Medical School, Boston, United States (04/2023) – **Poster**– **Winner of the Best Poster Award.**

4. **S. Busatto**, G. Morad, M. A. Moses. Brain-seeking extracellular vesicles derived from metastatic breast cancer cells modulate brain endothelial cell metabolism. Judah Folkman Research Day 2022. Boston Children's Hospital, Boston, United States (06/2022) – **Poster.**

5. **S. Busatto**, G. Morad, M. A. Moses. Extracellular vesicles released by brain-seeking cancer cells are key players in breast to brain metastasis. Judah Folkman Research Day 2021. Boston Children's Hospital, Boston, United States (06/2021) – **Poster.**

6. A. Zendrini, L. Paolini, **S. Busatto**, A. Radeghieri, M.H. Wauben, M. van Herwijnen, P. Nejsum, A. Borup, A. Ridolfi, C. Montis, P. Bergese. The CONAN assay: purity grade and concentration of EV microliter formulations by colloidal

nanoplasmonics. ISEV2020. Symposium Session 13: EV characterization (07/2020) – **Poster. Winner of the outstanding Oral Abstract Award.**

7. P. Bergese, D. Berti, M. Brucale, **S. Busatto**, S. Federici, C. Montis, L. Paolini, A. Radeghieri, A. Ridolfi, A. Salvatore, F. Valle, A. Zendrini. Projecting extracellular vesicles in two dimensions. 1st EVIta Symposium, Palermo, Italy (11/2019) – **Poster.**

8. L. Paolini, M. E. Forcella, P. Fusi, M. Brambilla, M. Frattini, F. Orizio, L. Paolini, **S. Busatto**, A. Radeghieri, R. Bresciani, P. Bergese, E. Monti. Active enzyme sialidase NEU3 is present on small extracellular vesicle surface. 1st EVIta Symposium, Palermo, Italy (11/2019) – **Poster.**

9. **S. Busatto**, S. Walker, P. Bergese, J. Wolfram. Extracellular vesicles secreted from braintropic cells display distinct interactions with brain endothelial cells. The first International Society of Extracellular Vesicles & Metastasis Research Society joint extracellular vesicle conference focused on cancer, Nashville, United States (08/2019) – **Poster.**

10. S. Walker, A. Pham, **S. Busatto**, J. Wolfram. Open Science Success in Community Outreach. SAGE assembly. Seattle, United States (07/2019) – **Poster.**

11. **S. Busatto**. Brain-Tropic Extracellular Vesicles for Treatment of Brain Tumors. New York Academy of Sciences STEM Exchange: Research and Career Symposium, New York, United States (8/2018) – **Poster** (selected as **one of 50 students from around the world**).

12. V. Porter, S. Singh, P. Che, **S. Busatto**, J. Wolfram, E. Brisbois and S.Seal. Nanoceria-integrated biopatch to accelerate wound healing. University of Central Florida Poster Session, Orlando, Florida, United States (8/2018) – **Poster.**

13. P. Bergese, **S. Busatto**, A. Zendrini, A. Radeghieri, D. Berti, C. Montis, A. Salvatore, F. Valle, Y. Gerelli. Projecting extracellular vesicles in two dimensions. 2018 ISEV Workshop membranes and extracellular vesicles**.** Baltimore, Maryland, United States (3/2018) – **Poster.**

14. L. Paolini, N. Bontempi, A. Radeghieri, A. Castelli, A. Zendrini, **S. Busatto**, E. Monti, I. Alessandri, P. Bergese. Probing nanosized extracellular vesicle (EV) populations by surface enhanced Raman spectroscopy (SERS). ISEV Annual International Meeting, Toronto, Canada (5/2017) – **Poster.**

Participant

1. **S. Busatto.** Center for Immunotherapeutic Transport Oncophysics (CITO). Spring Symposium: The physics of cancer immunotherapy. Houston Methodist Research Institute, Houston, Texas, United States (5/2018) – **Participant.**

PROFESSIONAL MEMBERSHIPS

2025 – present	Netherlands Society of Extracellular Vesicles (NLSEV) member
2017 - present	International Society of Extracellular Vesicles (ISEV) member
2020 - present	New York Academy of Sciences member
2019 - present	American Association for Cancer Research (AACR) member
2021- present	Metastasis Research Society (MRS) member

JOURNAL REVIEWER (https://publons.com/author/1386304/sara-busatto#profile)

Invited reviewer

1. Nature Nanotechnology
2. Advanced Science
3. Journal of Nanomedicine

4. Scientific Reports
5. Journal of Pharmacology and Clinical Toxicology
6. Molecular Therapy
7. ACS Nano
8. Journal of Nanobiotechnology
9. Molecular Therapy
10. Biochimica and Biophysica Acta
11. Scientific Reports
12. Frontiers in immunology
13. Molecular Therapy – Methods and Clinical Developments
14. Nano Select
15. Acta Biomaterialia

News articles

Sara Busatto – A different kind of Florida visitor. Advancing the Science Mayo Clinic Medical Science Blog, United States (4/2018) https://advancingthescience.mayo.edu/2018/04/09/sara-busatto-a-different-kind-of-florida-visitor/

Sara Busatto is not in Florida for the Flamingos, Mayo Clinic News Center, United States (4/2018) http://newsletters.mayo.edu/newscenter/Article.aspx?contentID=DOCMAN-0000174463 (internal link)

WEBSITES and SOCIAL MEDIA

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